

Unit 12 - Data Displays

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The following B.E.S.T Standards will be covered in this unit:

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

- *Clarification 1: Instruction includes discussions regarding the strengths and weaknesses of each data display.*
- *Clarification 2: Numerical univariate includes histograms, stem-and-leaf plots, box plots and line plots; numerical bivariate data includes scatter plots and line graphs; categorical univariate includes bar charts, circle graphs, line plots, frequency tables, and relative frequency tables; and categorical bivariate includes segmented bar charts, joint frequency tables and joint relative frequency tables.*
- *Clarification 3: Instruction includes the use of appropriate units and labels and, where appropriate, using technology to create data displays.*

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate and interpret the different components and quantities in the display.

MA.912.DP.1.3 Explain the difference between correlation and causation in the contexts of both numerical and categorical data.

MA.912.DP.1.4 Estimate a population total, mean or percentage using data from a sample survey; develop a margin of error through the use of simulations.

- *Clarification 1: Within the Algebra 1 course, the margin of error will be given.*

Unit 12, Lesson 1: Displaying Univariate Data



Benchmarks

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate and interpret the different components and quantities in the display.



Learning Targets

- ☐ I can understand how various displays can be used to represent a univariate categorical data set.
- ☐ I can select an appropriate data display to represent numerical or categorical univariate data.



12.1.1 Warm-Up: Bell Work

To choose an appropriate graphical representation for data, the first step is to determine if the data is **numerical** (data values are numbers) or **categorical** (data values fall within categories).

1. Complete the table.

Possible Question	Possible Responses	Numerical or Categorical
What is your favorite color?	red, orange, yellow, green	☒ categorical ☐ numerical
How many people are in your family?	2, 3, 4, 5 ...	☐ categorical ☒ numerical
How old are you?	11, 12, 13 ...	☐ categorical ☐ numerical
How tall are you?		☐ categorical ☐ numerical
What is your favorite sport?		☐ categorical ☐ numerical
	watermelon, apple, pear, orange	☐ categorical ☐ numerical



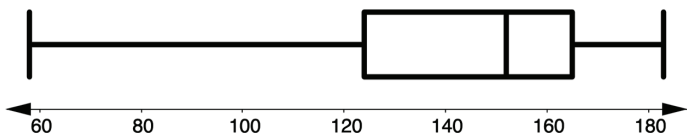
12.1.2 Refresher: Choosing the Best Graphical Representation for Univariate Numerical Data

Four graphical representations that can be used to display **numerical data** that is **univariate** are box plots, histograms, line plots, and stem-and-leaf plots. Each of these graphical representations is shown in the table.

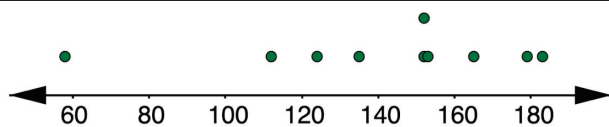
Univariate data is data that measures one characteristic of a population.

Graphical Representations for Univariate Quantitative Data

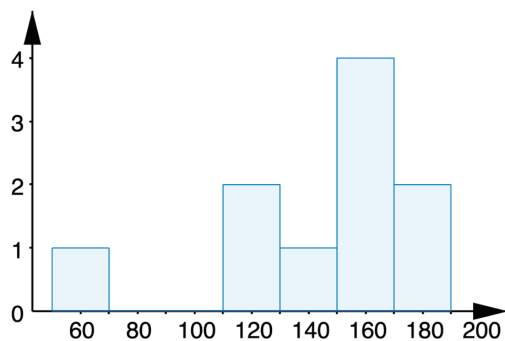
Box Plot



Line Plot



Histogram



Stem-and-Leaf Plot

Stem	Leaf
5	8
6	
7	
8	
9	
10	
11	2
12	4
13	5
14	
15	2, 2, 3
16	5
17	9
18	3

Key: 5|8 = 58

Work with your partner to complete the following.

1. What do you notice about the displays?

2. Complete the table by checking the displays that match each attribute.

Attribute	Box Plot	Line Plot	Histogram	Stem and Leaf Plot
Shows the distribution of the data set	☐	☐	☐	☐
Shows the shape of the data distribution	☐	☐	☐	☐
Data is displayed in intervals (also called bins)	☐	☐	☐	☐
Displays each individual data value	☐	☐	☐	☐
Displays a 5-number summary of the data set	☐	☐	☐	☐

3. Camilla wants to create a graphical display that represents the typing speeds of the 425 students at her school.

a. Which display(s) will best display the data?

- ☐ box plot
- ☐ line plot
- ☐ histogram
- ☐ stem-and-leaf plot

b. Explain your reasoning.

c. Camilla says, "Making a line plot of this data would take a lot of time and it would be hard to make sure each dot is in the right location on the number line, since speed can be given in fractions of seconds. I'd have to make 425 dots to represent each student!" Explain whether you agree with Camilla that a line plot would not best display the data.

4. Dallas records the highest temperature from each month over the past year in Immokalee, Florida.

$\{76, 78, 81, 85, 90, 92, 92, 93, 91, 87, 82, 78\}$

a. Which display(s) will best display the data?

- ☐ box plot
- ☐ line plot
- ☐ histogram
- ☐ stem-and-leaf plot

b. Explain your reasoning.

c. Dallas decides to change his data set to include the average weekly high temperature. Explain whether this changes the display he should use.

5. Basil recorded the weights, rounded to the nearest pound, of 15 dogs at a pet show.

7, 10, 12, 22, 28, 59, 62, 65, 72, 79, 85, 98, 100, 150, 163

- a. Explain which display would be better for the data, a line plot or a stem-and-leaf plot.

- b. Basil also recorded the weights, rounded to the nearest pound, of 15 cats at the pet show.

4, 4, 5, 7, 7, 8, 8, 8, 9, 10, 11, 12, 13, 13, 14

Explain which display would be better for the data, a line plot or a histogram.

- c. He updates his data to show the weights to the nearest tenth.

4.1, 4.3, 5.2, 7.6, 7.9, 8.0, 8.2, 8.7, 9.3, 10.5, 11.2, 12.8, 13.4, 13.5, 14.6

Explain which display would be better for the data, a line plot or a histogram. If your recommendation is different from Part B, explain why.

6. Ms. Rallen asked her students how many pets they each have, and wants to use a graphical representation that will allow her students to see the overall shape of the data and give them enough information to calculate the 5-number summary.

- a. Which graphical representations should Ms. Rallen consider?

- ☐ box plot
- ☐ line plot
- ☐ histogram
- ☐ stem-and-leaf plot

- b. Explain your reasoning.



12.1.3 Collaboration: Graphical Representations of Univariate Categorical Data

Graphical representations that can be used to display univariate categorical data include **bar graphs**, **circle graphs**, **line plots**, **frequency tables**, and **relative frequency tables**.

Categorical data is often collected in a frequency table before creating another display.

Consider the data summarized in the frequency table and relative frequency table below, collected from a survey on how students get to school.

Frequency Table

Mode of Transportation	Frequency
Bus	129
Car	71
Bike	65
Walk	35

Relative Frequency Table

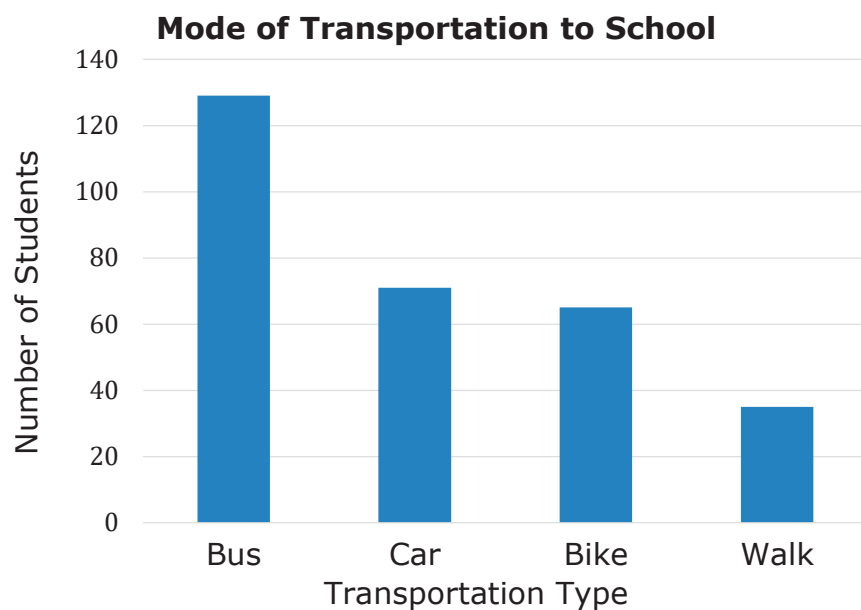
Mode of Transportation	Relative Frequency
Bus	43%
Car	24%
Bike	22%
Walk	11%

1. Summarize the similarities and differences between the two displays.

Similarities	Differences

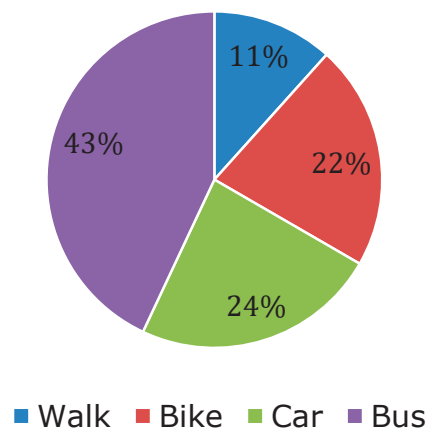
2. How was the data from the frequency table used to construct the relative frequency table?

The vertical bar graph and circle graph show the same data.



The circle graph shows the same data.

Mode of Transportation to School



3. Complete the statements.

The bar graph visually shows

- ☐ how each part relates to the whole.
- ☐ how each part compares to the other parts.

The circle graph shows

- ☐ how each part relates to the whole.
- ☐ how each part compares to the other parts.

The data from the frequency table was used to

construct the

- ☐ bar graph.
- ☐ circle graph.

The data from the relative frequency table was used to

construct the

- ☐ bar graph.
- ☐ circle graph.

4. Similar to a line plot (dot plot) for quantitative data, a line plot can also be used for categorical data by listing each category on the x -axis. Would a line plot be a good choice for displaying this data? Explain your reasoning.



12.1.4 Wrap-Up: Cool Down

1. My favorite graphical representation to display

numerical univariate data is

- box plot.
- line plot.
- histogram.
- stem and leaf.

It is my favorite because ... _____

_____.

2. My favorite graphical representation to display

univariate categorical data is a

- bar graph.
- circle graph.
- line plot.
- frequency table.

It is my favorite because ... _____

_____.



Unit 12, Lesson 2: Creating Displays for Numerical Data Sets



Benchmarks

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, and whether it is univariate or bivariate and interpret the different components and quantities in the display.



Learning Targets

- ☐ I can distinguish between univariate and bivariate data, including which representations can be used to display each type of data set.
- ☐ I can create a graphical display for univariate numerical data.
- ☐ I can create a graphical display for bivariate numerical data.



12.2.1 Warm-Up: Classifying Data

When determining the best display for a data set, the first step is to determine the type of data you are being given.

Data can be classified as categorical or numerical, and univariate (one variable) or bivariate (two variables).

1. Identify the type of data for each example in the table. The Pew Research Center collected data from a sample of 743 teens to analyze trends in social media and technology use.

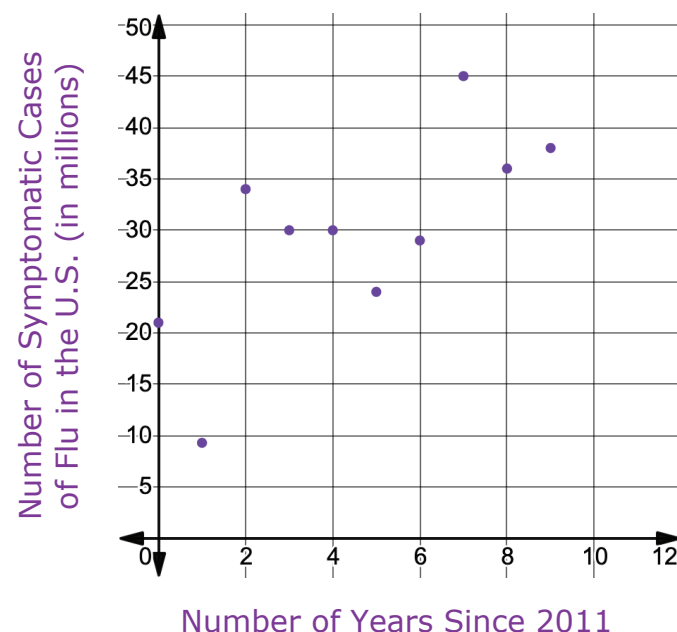
Example	Categorical or Numerical?	Univariate or Bivariate?
a. The number of people you consider close friends	☐ Categorical ☐ Numerical	☐ Univariate ☐ Bivariate
b. The number of people you consider close friends and the time spent on social media per day	☐ Categorical ☐ Numerical	☐ Univariate ☐ Bivariate
c. Your most used social media site (Twitter, Instagram, Facebook, Snapchat, YouTube, or Reddit)	☐ Categorical ☐ Numerical	☐ Univariate ☐ Bivariate
d. Your most used social media site (Twitter, Instagram, Facebook, Snapchat, YouTube, or Reddit) and what effect you would say social media has had on people your age (mostly positive, mostly negative, neutral)	☐ Categorical ☐ Numerical	☐ Univariate ☐ Bivariate



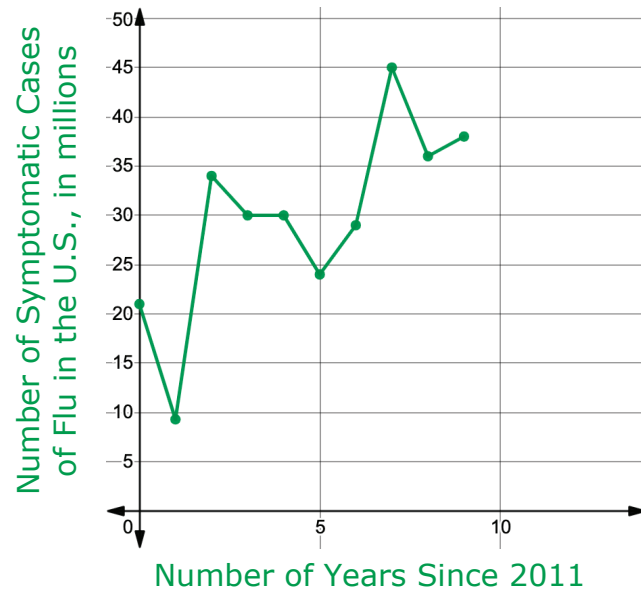
12.2.2 Refresher: Displays for Bivariate Numerical Data

Bivariate numerical data can be represented in different ways, including **scatter plots** and **line graphs**.

A scatter plot of the data on symptomatic cases of the flu in the U.S. is shown, where y is the number of symptomatic cases of the flu in the U.S., in millions, and x is the number of years since 2011.



The same data can be represented using a **line graph**.



1. Discuss with your partner the similarities and differences between the scatter plot and the line graph.

Complete the table.

Similarities Between Scatter Plots and Line Graphs	Differences Between Scatter Plots and Line Graphs

2. If data was collected on the number of symptomatic flu cases for each state for every year since 2011, which display would you choose to represent the data? Explain your reasoning.
3. What do you notice about the relationship between the number of symptomatic flu cases and years since 2011?



Both scatter plots and line graphs can be used to observe the relationship between two variables in a set of bivariate data. While overall trends can be seen in both graphs, line graphs allow for closer analysis from one point to another in the data set. Scatter plots are useful for larger sets of data.



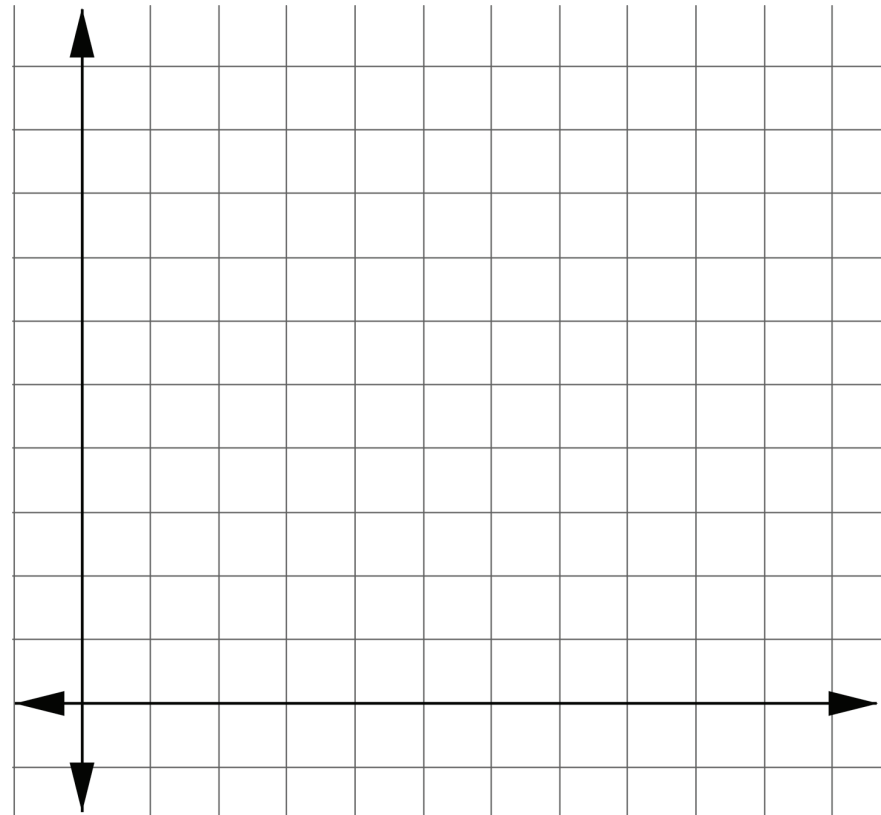
12.2.3 Try It: Creating Displays for Numerical Data

1. The prices and size, in carats, of 20 different diamonds are shown in the table.

Weight (carats)	Actual Price (\$)
1	3,772
1	4,221
1	4,032
1	5,385
1.05	3,942
1.05	4,480
1.06	4,511
1.2	5,544
1.3	6,131
1.32	5,872
1.41	7,122
1.5	7,474
1.5	5,904
1.59	8,706
1.61	8,252
1.73	9,530
1.77	9,374
1.85	8,169
1.9	9,541
2.04	9,125

- a. The data shown in the table is ☐ numerical
and ☐ univariate.
☐ categorical
☐ bivariate.
- b. The data shown in the table should be displayed in
a ☐ scatter plot
☐ line graph because ...

- c. Create a display to represent the relationship between size and price of a diamond.



2. Laverne and Rocco planned to travel the greatest distance between any two points in the United States — from Log Point, Elliot Key, Florida to Kure Atoll, Hawaii — a distance of 5,859 miles.

They took a ferry from Log Point to Miami, then drove to Los Angeles, California, where they took a plane to Hawaii. They drove anywhere from 5-8 hours a day, and kept track of how many miles they traveled per day on their trip. This information is shown below.

{224, 294, 309, 317, 321, 339, 355, 361, 383, 456, 2500}

- a. Complete the statement.

The data set is

- ☐ numerical
- ☐ categorical

and

- ☐ univariate.
- ☐ bivariate.

- b. Select all of the displays that **can** be used to display the data.

- ☐ Dot plot
- ☐ Histogram
- ☐ Box plot
- ☐ Stem-and-leaf
- ☐ Bar graph
- ☐ Circle graph
- ☐ Scatter plot

- c. Select all of the displays that **are best** to display the data.

- ☐ Dot plot
- ☐ Histogram
- ☐ Box plot
- ☐ Stem-and-leaf
- ☐ Bar graph
- ☐ Circle graph
- ☐ Scatter plot

- d. Create an appropriate graphical representation for the data.

- e. If the value 2500 is removed from the data set, which display would you create to represent the data? Explain your reasoning.



3. Basil recorded the weights, rounded to the nearest pound, of 15 cats at a pet show.

4.1, 4.3, 5.2, 7.6, 7.9, 8.0, 8.2, 8.7, 9.3, 10.5, 11.2, 12.8, 13.4, 13.5, 14.6

- a. Create an appropriate graphical display for the data. Do not use the same type of display you created in the previous problem.

- b. What other displays could be used to display his data?



12.2.4 Bring it Together: Choosing the Best Graphical Representation for Numerical Data

Univariate, numerical data:

Line Plots (Dot Plots)



- Best for small to moderate sets of data
- Shows the shape and distribution of data
- Shows individual data values



Histograms

- Best for larger sets of data and data over a large range,
- Can be used for continuous data
- Individual data values are not shown, only frequency over intervals
- Shows shape and distribution of the data



Stem-and-Leaf Plots

- Best for small data sets
- Shows the shape and distribution of data
- Displays individual data values in a more compressed way than a dot plot



Box Plots

- Best for large sets of data
- Displays variation of the data
- Individual data values are not shown
- Shows the 5-number summary of the data set

Numerical, bivariate data:



Scatter plots

- Best for large sets of data
- Helpful for looking at trends over a large data set
- Useful for comparing two quantities without thinking of either one as the independent or dependent variable



Line Graphs

- Best for smaller sets of data
- Helpful for looking at small changes over time



12.2.5 Wrap-Up: Reflection

Complete each statement based on today's lesson.

1. Today I am still unsure about ...

2. One question I still have is ...

Unit 12, Lesson 3: Interpreting Data Distributions for Numerical Data



Benchmarks

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate and interpret the different components and quantities in the display.



Learning Targets

- ☐ I can interpret data distributions from an appropriate display for univariate, numerical data.
- ☐ I can interpret data distributions from an appropriate display for bivariate, numerical data.

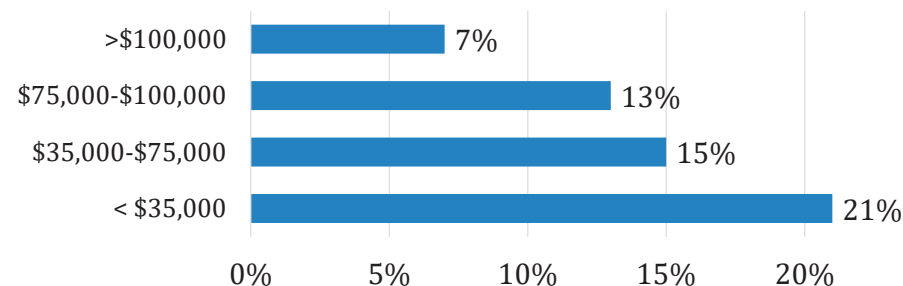


12.3.1 Warm-Up: Smoking and Income

In 2019, about 14% of U.S. adults smoked cigarettes, a decline from about 20% in 2005.

The bar graph shows the percent of smokers by household income.

Smokers by Household Income



1. What do you notice?

2. What do you wonder?



12.3.2 Refresher: Measures of Center and Variation



Measures of center, such as **mean** and **median**, are used to describe a typical value in the data set.



Measures of variation, such as **range** and **interquartile range**, are used to describe how “spread out” the data is.



The **mean** is the arithmetic average of a set of numbers. It is a measure of center, or central tendency.



The **median** is the middle of an ordered list of the values. If the list has an odd number of values, it is the middle value of that list. If the list has an even number of values, it is the average of the two middle values. It is a measure of center, or central tendency.



The **range** is the difference between the highest data value and the lowest data value in a data set. It is a measure of variation.



A measure of variation in a set of numerical data, the **interquartile range (IQR)** is the distance between the first and third quartiles of the data set.

1. Use the word bank to complete the following. Not all terms will be used.

mean median mode range
interquartile range quartile



When a distribution is bell-shaped, the _____ is the preferred measure of center and the _____ is the preferred measure of variation to describe the distribution. However, the context must be considered.

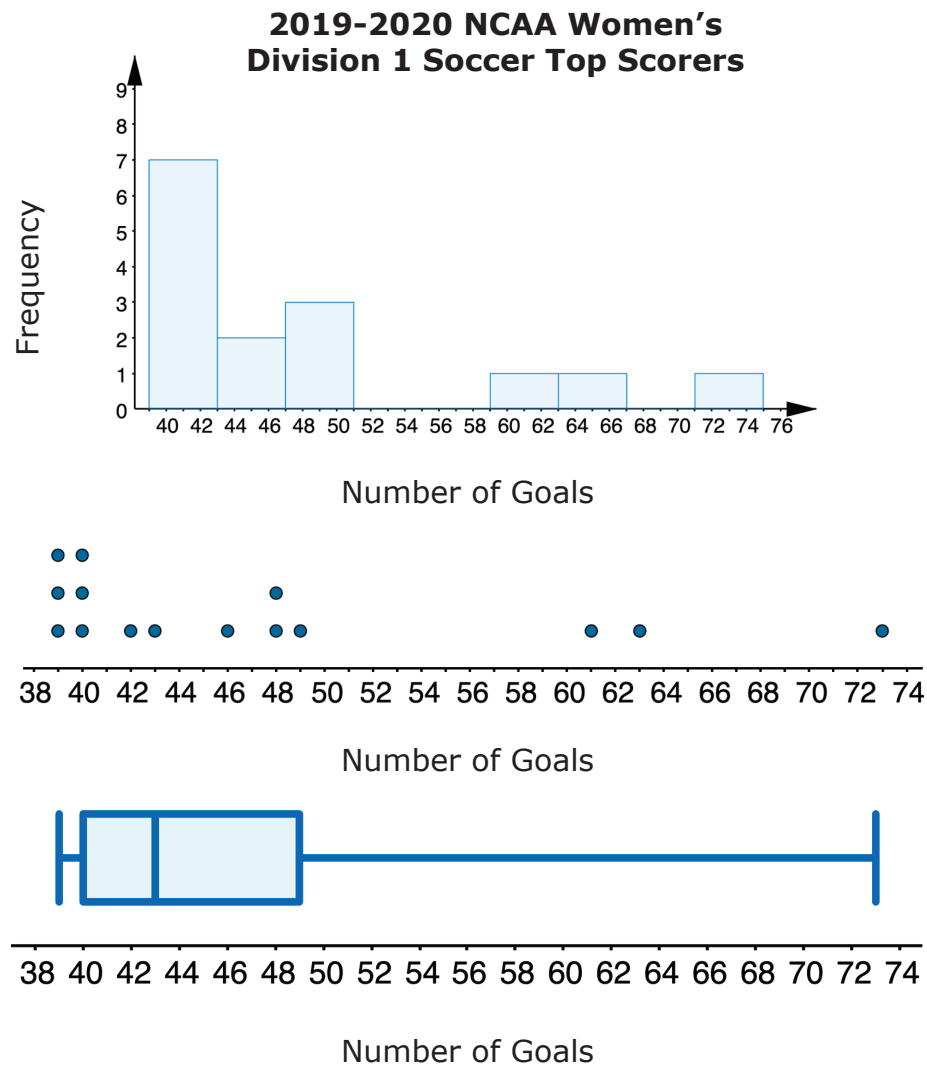


When outliers are present or a distribution is skewed, the _____ is the preferred measure of center and the _____ is the preferred measure of variation to describe the distribution. However, the context must be considered because outliers may be important in understanding the data.



12.3.3 Guided Practice: Comparing Distributions

1. Shown are a histogram, line plot, and box plot of the number of goals scored by the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season.



- a. The ☐ histogram ☐ line plot ☐ box plot can be used to determine the mean.
- b. The ☐ histogram ☐ line plot ☐ box plot can be used to determine the median.
- c. What is the appropriate measure of center for the data set?
- d. Which display would you prefer to use to determine the median?
- e. Interpret the median of the data set.

f. The ☐ histogram ☐ line plot ☐ box plot can be used to determine the range.

g. The ☐ histogram ☐ line plot ☐ box plot can be used to determine the interquartile range.

h. What is the appropriate measure of spread for the data set?

i. Which display would you prefer to use to determine the interquartile range?

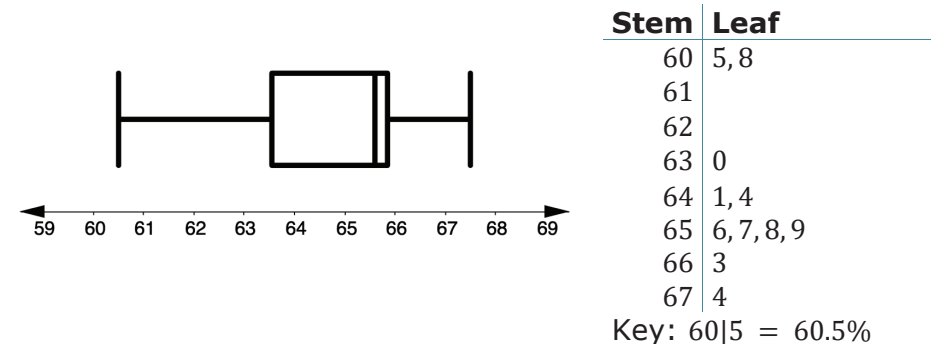
j. Interpret the interquartile range.



12.3.4 Collaboration: Univariate vs. Bivariate Data

- Tom Brady is considered by most to be one of the best NFL quarterbacks of all time. His pass-completion percentages from 2009 to 2019 are shown in the box plot and stem-and-leaf plot.

Tom Brady's Completion Percentages 2009-2019

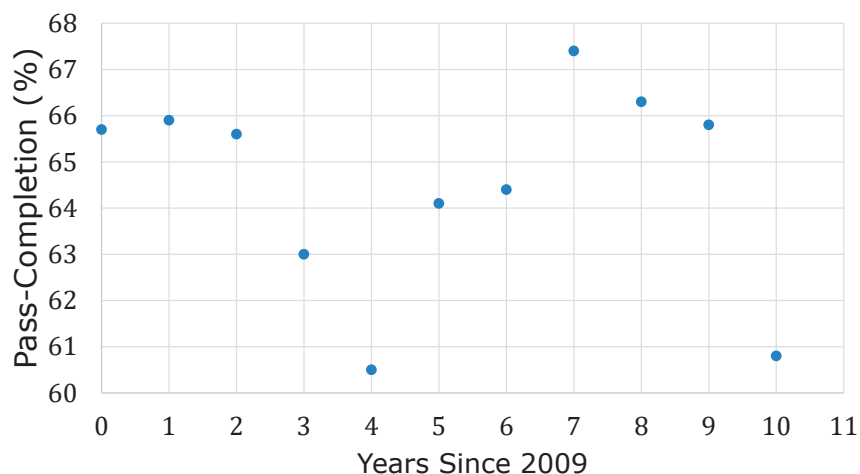


- The ☐ mean ☐ median best summarizes the center of the data. The ☐ mean ☐ median is not the best measure of center because it is influenced by the cluster of values, ☐ 60.5% and 60.8%, ☐ 66.3% and 67.4%, that are far from the rest of the data values.
- The ☐ IQR ☐ range best summarizes the variation of the data.

2. Tom Brady's pass-completion percentages from 2009 to 2019 are listed in the table and shown in the scatter plot.

Year	2009	2010	2011	2012	2013	2014
%	65.7	65.9	65.6	63	60.5	64.1
Year	2015	2016	2017	2018	2019	
%	64.4	67.4	66.3	65.8	60.8	

Tom Brady's Pass-Completion 2009-2019



- a. The scatter plot displays the relationship between

two ☐ numerical ☐ categorical variables. Another display that

could be created for this ☐ univariate ☐ bivariate data set

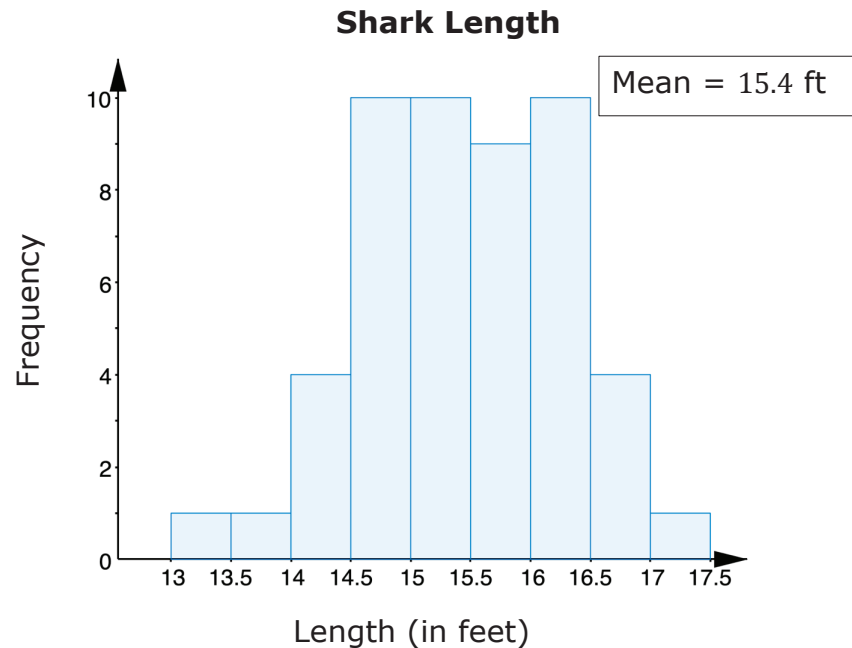
is a ☐ line graph. ☐ histogram.

- b. What is one observation you can make from the scatter plot that you could not from the displays summarizing only the pass-completion percentages?



12.3.5 Guided Instruction: Summarizing Data

1. The histogram summarizes the lengths, in feet (ft), of a group of adult female sharks.



- a. Describe the shape of the distribution.
- b. Interpret the mean of the distribution by completing the following statement.

The typical length of an adult female shark is _____.

- c. Interpret the variation in shark length.

Another measure of variation that summarizes bell-shaped numerical data is **standard deviation**.

The **standard deviation** of a data set is the typical variation of each value to the mean. The larger the standard deviation, the more spread out the data is.

- d. The standard deviation of shark length is 0.75 ft. Interpret the standard deviation.
- For this sample of adult female sharks, the typical deviation from the mean of _____ is _____. This group of sharks have lengths that are relatively similar.



When summarizing data, both a measure of center and a measure of variability should be used to provide a summary of both what is typical of the data (center) *and* how far the data is spread from that value (variability).

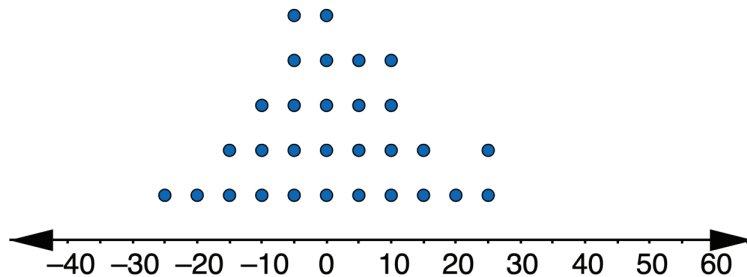
- When using the **mean** to summarize the center, **standard deviation** should be used to describe the spread when the data is bell-shaped.
- When using the **median** to summarize the center, **interquartile range** should be used to describe the spread.



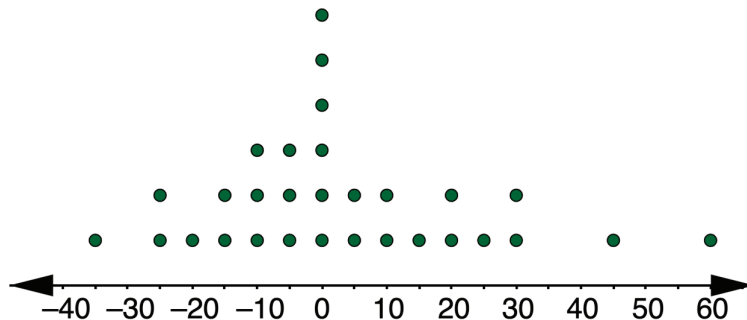
12.3.6 Try It: Summarizing Data

1. The dot plots display the arrival times of 30 flights for two different airlines.

Arrival Times of Airline P



Arrival Times of Airline Q



A negative number represents the number of minutes the flight arrived before its scheduled time.

A positive number represents the number of minutes the flight arrived after its scheduled time.

A zero indicates that the flight arrived at its scheduled time.

- a. Which airline do you expect to have a larger standard deviation?

- b. Based on these data, from which airline would you choose to buy your ticket? Use measure(s) of center and variation to justify your choice.



12.3.7 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson?



Unit 12, Lesson 4: Displaying and Interpreting Univariate Categorical Data



Benchmarks

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate and interpret the different components and quantities in the display.



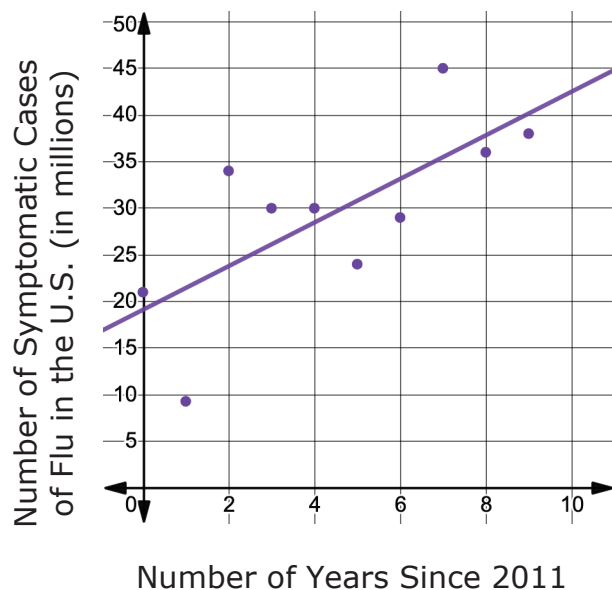
Learning Targets

- ☐ I can create an appropriate graphical display for univariate categorical data.
- ☐ I can interpret a graphical display for univariate categorical data.



12.4.1 Warm-Up: Line of Fit on a Scatter Plot

A scatter plot of the data on symptomatic cases of the flu in the U.S. is shown. The line of fit, $y = 2.34x + 19.1$, is included on the scatter plot, where y is the number of symptomatic cases of the flu in the U.S., in millions, and x is the number of years since 2011.



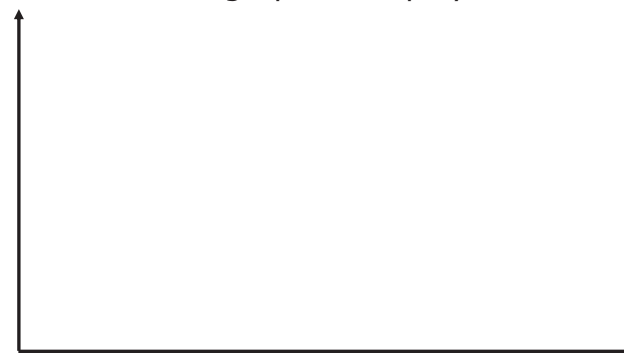
1. Interpret the slope of the line of fit.
2. Interpret the y -intercept of the line of fit.



12.4.2 Guided Instruction: Displays for Categorical Data

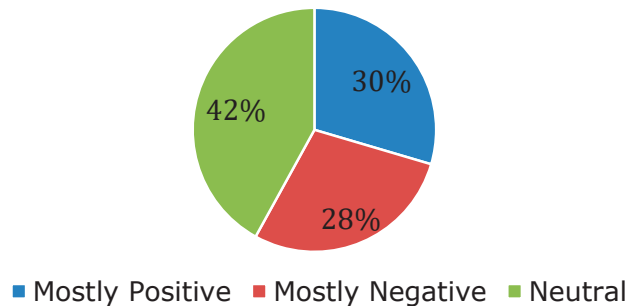
A 2018 Pew Research Center survey of American teenagers was conducted to analyze trends in social media and technology.

1. The results for 741 teens who use social media, when asked what effect they believe social media has had on people their age (mostly positive, mostly negative, neutral), is summarized.
 - 219 said the effect was mostly positive
 - 211 said the effect was mostly negative
 - 311 said the effect was neutral
- a. Which of the following display(s) can be used to represent the data? Select all that apply.
 - ☐ Box plot
 - ☐ Histogram
 - ☐ Circle graph
 - ☐ Frequency table
 - ☐ Relative frequency table
 - ☐ Bar graph
 - ☐ Scatter plot
- b. Create a bar graph to display the data.



2. The same data is summarized in the circle graph.

Teen Views on the Effect of Social Media on People Their Age



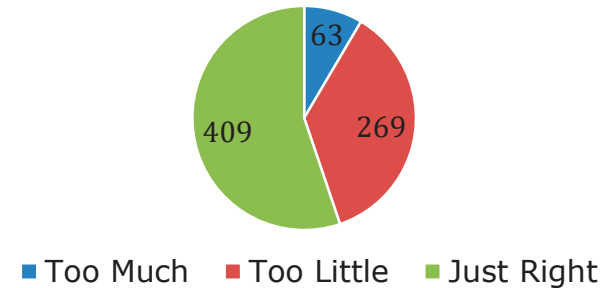
The percentage of teens who describe social media as having a mostly positive effect on people

their age is ☐ less than ☐ equal to ☐ greater than the percent of students who believe it has had a negative effect.

On the same survey, data was also collected on how much time they believe they spend with their friends in person (too much, too little, or about the right amount).

3. The data is summarized in the circle graph.

Time Spent With Friends in Person



What percentage of participants feel that they spend too little time with their friends in person?

4. What do you know about the relationship between time spent with friends in person and perceived effect of social media?

5. To analyze the relationship between how much time they spend with their friends and their perceived effect social media has on people their age, an appropriate

display for ☐ numerical, ☐ categorical, ☐ univariate, ☐ bivariate

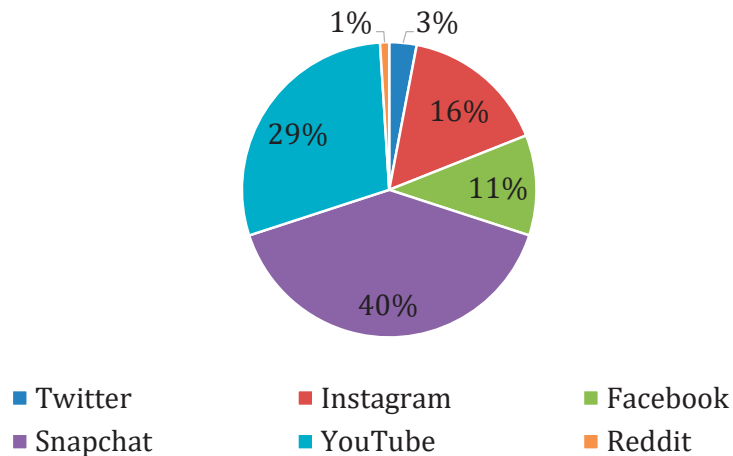
data should be created.



12.4.3 Your Turn!

- In the 2018 Pew Research Center survey of American teenagers, 634 respondents used more than one social media site. Data on their most used site is summarized in the circle graph.

Most Used Social Media Site



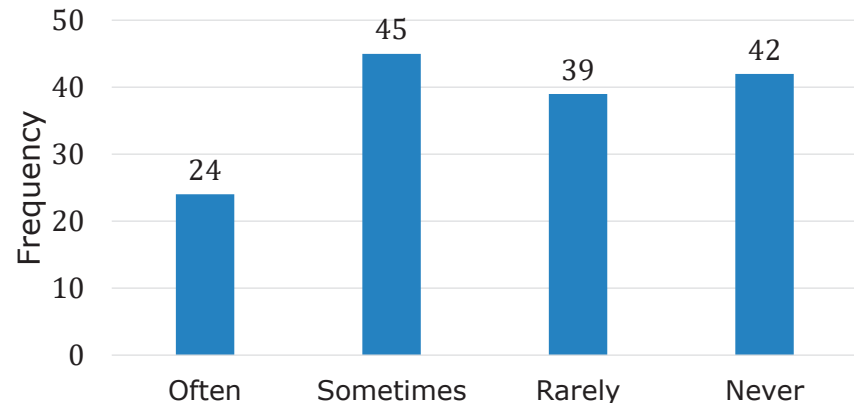
- What was the most used social media site?
- What was the least used social media site?
- How many respondents said Instagram was their most used site?



12.4.4 Wrap-Up: Error Analysis

- A survey of U.S. teens asked participants how often they post selfies. The results are summarized in the bar graph.

How Often Teens Post Selfies



Rayah analyzed the bar graph and concluded that the majority of U.S. teens in the sample post selfies sometimes or often since the percent who post often plus the percent who post sometimes is more than 50%.

- Explain the error in Rayah's interpretation.
- Find the percent of U.S. teens in the sample who post selfies sometimes or never.

Unit 12, Lesson 5: Correlation vs. Causation



Benchmark

MA.912.DP.1.3 Explain the difference between correlation and causation in the contexts of both numerical and categorical data.



Learning Target

- ☐ I can explain the difference between correlation and causation.



12.5.1 Warm-Up: Which One Doesn't Belong?

1. Which one doesn't belong?
 - A. The number of miles driven and the amount of gas left in the tank
 - B. The number of social media posts last summer and the number of movies seen this summer
 - C. Time spent exercising and calories burned
 - D. The number of women diagnosed with breast cancer and the presence of BRCA1 or BRCA2, the gene mutations that raise risk for the disease
2. Explain your reasoning.



12.5.2 Collaboration: What Affects Association?

The following data was collected to explore the relationship between neighborhood income and access to organic vegetables at local grocery stores.

Average Household Income	Organic Vegetables Offered	Average Household Income	Organic Vegetables Offered
\$71,186	36	\$50,252	65
\$34,234	4	\$53,945	50
\$71,186	28	\$59,072	35
\$48,760	31	\$49,437	36
\$78,096	78	\$72,080	28
\$40,506	14	\$108,486	95
\$38,166	12	\$70,530	46
\$50,398	18	\$57,199	29
\$49,437	38	\$78,176	73
\$66,073	84	\$78,288	53
\$86,566	61	\$86,566	86
\$78,176	56	\$84,181	68
\$59,154	62	\$84,181	56
\$50,252	44	\$78,176	85
\$48,364	26	\$84,181	86
\$56,274	29	\$135,547	93
\$41,318	15	\$92,946	82
\$125,145	95	\$77,894	96
\$65,911	18		

1. The data shown in the table is

- numerical,
- categorical,

- univariate
- bivariate

data. To explore the relationship

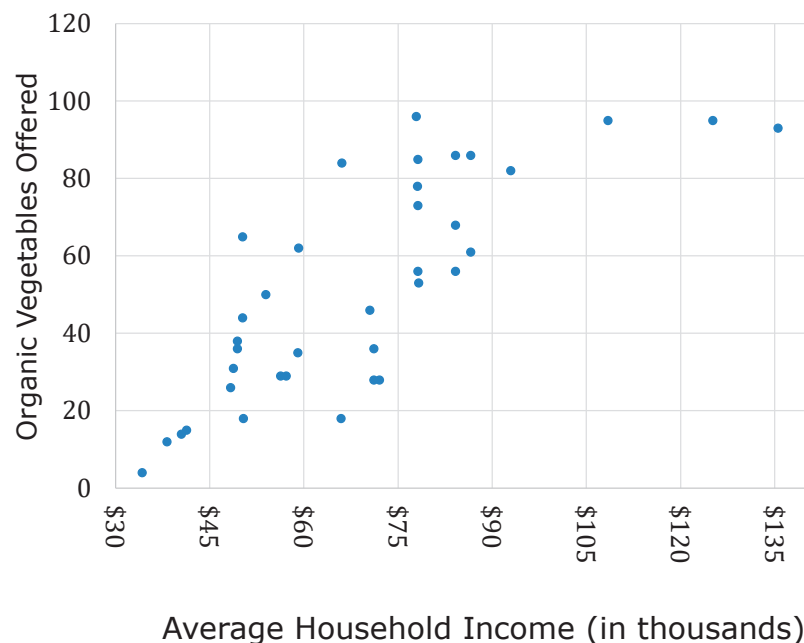
visually, a

- scatter plot
- histogram

can be created.

A scatter plot of the same data is shown.

Income and Food Access



2. Describe the relationship between average household income and access to organic vegetables.

3. The data has a correlation coefficient of 0.79.
 - a. Explain what this means in terms of the data.
 - b. Discuss with your partner whether you think this means that having a high household income causes more organic food to be available. Summarize your discussion.
4. What other factors could contribute to the association between income and organic food availability?



12.5.3 Exploration: Does Association Mean Causation?

1. There is a strong positive association between the amount of fire damage (in dollars) and the number of firefighters on the scene of a fire.
 - a. Does having more firefighters on the scene cause greater fire damage?
 - b. What other factor may contribute to the association between these variables?
2. A 2011 article in the *Washington Post* had the following headline.

Soda boosts violence among teens, study finds

The study found that there is a positive correlation between soft-drink consumption and carrying a gun or knife and violence.

- a. Does this mean that if you drink more soda you increase your chance of violence?

- b. What other factors could contribute to this association?

3. A study published in the *Archives of Internal Medicine* found that regular chocolate consumption was related to lower body mass index (BMI).
- a. Does this mean that eating chocolate will make you thin?
- b. What other information would be helpful to evaluate that claim?



Correlation does not imply causation!

If there is a correlation, or association, between two variables, it does not mean that one causes the other.

Two variables can be correlated without having a causal relationship.



12.5.4 Guided Instruction: Causation

Establishing a correlation, or relationship, between two variables can be straightforward. For numerical bivariate data, a correlation coefficient can be computed and interpreted, or displays such as scatter plots can be created and analyzed.

If there is a correlation between two variables, causation **might** exist as well.

However, establishing causation can be difficult.

1. Causation cannot be established by analyzing survey data. Why do you think survey data can't be used to establish causation?



Causation can only be concluded if a controlled study or experiment was performed. Causation applies to a relationship when an action **causes** the outcome.

2. In the recent development of a vaccine, half of the participants were given a placebo, or fake dose, and the other half received the true vaccine. Symptoms were measured 21 days after injection.

If the data shows evidence of an association between the vaccine and disease contraction, could this controlled experiment be used to establish causation?



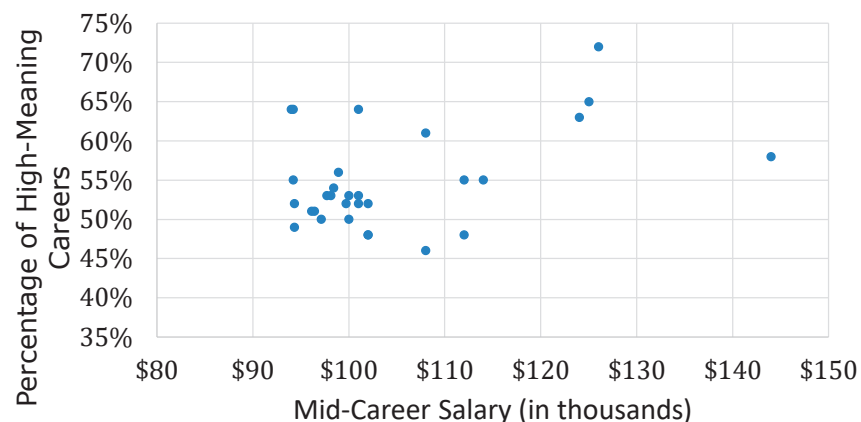
While the word 'correlation' refers to a relationship or association between two things, a correlation coefficient can only be used to quantify the linear relationship between two numerical variables.



12.5.5 Try It: Relationship Between Two Numerical Variables

1. The scatter plot summarizes the top 30 public universities based on the median mid-career salary of alumni who earned a bachelor's degree but not an advanced degree. The table also shows the percentage of bachelor's earning alumni with high-meaning careers — that is, alumni who say their work makes the world a better place (*PayScale*, n.d.; *U.S. News*, n.d.).

Mid-Career Salary vs. Percentage of High-Meaning Careers

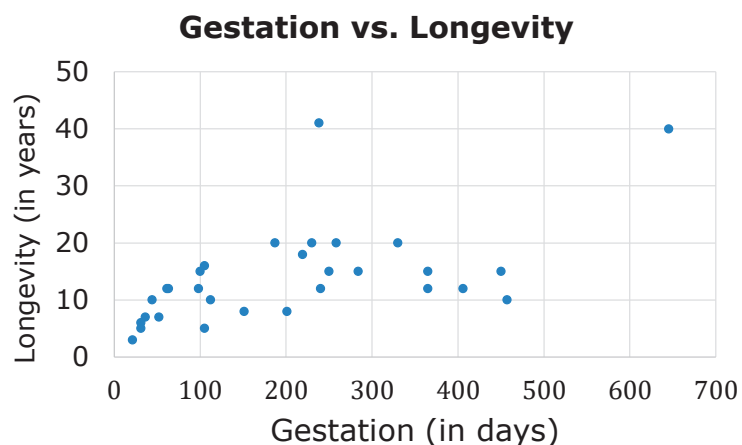


- a. Describe the relationship between mid-career salary and the percentage of alumni who say their work makes the world a better place.
- b. Why can't causation be established in this context?



12.5.6 Wrap-Up: Ticket Out the Door

1. The following scatter plot summarizes the average gestation period, in days, and average longevity, in years, for a sample of 30 animals as reported in *The World Almanac and Book of Facts* (McGeveran, 2006). A gestation period is the amount of time for which an animal is pregnant. Longevity refers to how long an animal lives.



- a. What does correlation mean in this context?
- b. Does the data provide evidence that a longer gestation period causes animals to live longer?

Unit 12, Lesson 6: Estimating for a Population



Benchmark

MA.912.DP.1.4 Estimate a population total, mean, or percentage using data from a sample survey; develop a margin of error through the use of simulation.



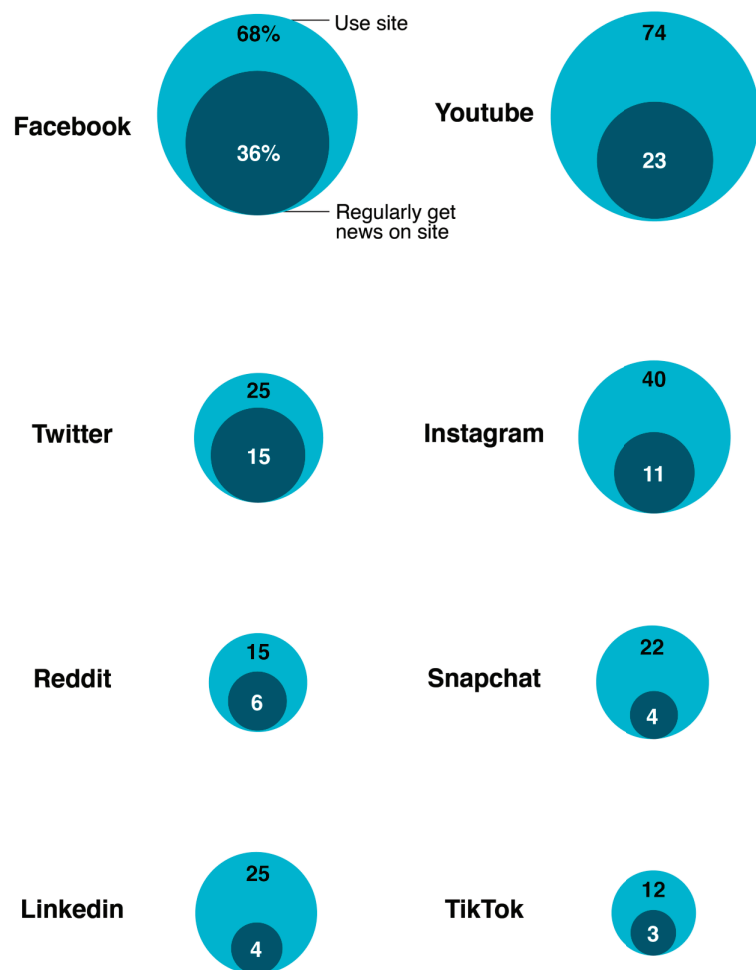
Learning Target

- ☐ I can use margin of error to estimate a population total, mean, or percentage using data.



12.6.1 Warm-Up: News Source

1. The graphic below summarizes data from a 2020 Pew Research Study. It shows the percent of U.S. adults who use each site and regularly get news on each site.



What do you notice? What do you wonder?



12.6.2 Guided Instruction: Margin of Error

In 2020, 9,220 U.S. adults were surveyed about their news use across social media platforms.

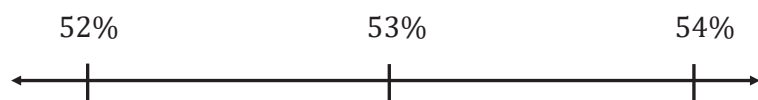
53% of those surveyed reported that they get their news from social media "often" or "sometimes."

1. If you were able to conduct a similar survey of U.S. adults, do you think exactly 53% would report that they get their news from social media "often" or "sometimes?" Explain your reasoning.
2. Can it be concluded that exactly 53% of all U.S. adults get their news from social media "often" or "sometimes?"
3. Do you think that the majority of U.S. adults get their news from social media sometimes or often?

4. Estimate the total number of U.S. adults who got their news from social media often or sometimes in 2020 if there were 256,662,010 U.S. adults at the time of the survey.

When a population percentage or mean is estimated from sample data, it is often helpful to give a range for the estimate to consider variability in the sample.

Based on the same sample data, it can be concluded that the population percentage is between 52% and 54%.

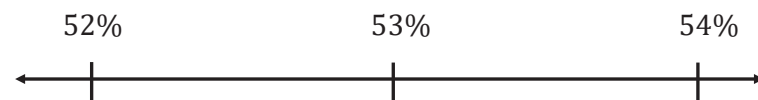


This interval of possible values for the estimate of the population percentage was created using a **margin of error**.

The **margin of error** quantifies the amount of random sampling error in the results of a survey.

In the sample data of 9,220 U.S. adults, the percentage who get their news from social media “often” or “sometimes” is 53% with a margin of error of $\pm 1\%$.

5. Estimate the range for the total number of U.S. adults who got their news from social media often or sometimes in 2020 if there were 256,662,010 U.S. adults at the time of the survey using the estimate of 53% with a margin of error of $\pm 1\%$.



$$0.52 \cdot 256662010$$

$$0.54 \cdot 256662010$$

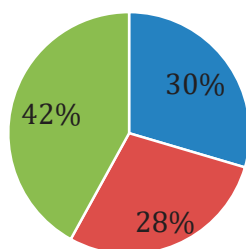
The total number of U.S. adults who get their news from social media sometimes or often is estimated to be between _____ and _____.



12.6.3 Guided Practice: Estimating for a Population

1. The results of 741 teens surveyed in a 2018 Pew Research Study are shown in the circle graph.

Teen Views on the Effect of Social Media on People Their Age



■ Mostly Positive ■ Mostly Negative ■ Neutral

- a. If there were 24.97 million teens in the U.S. population at the time of the survey, what is the estimated number who believed the effect of social media on people their age had been mostly positive?
- b. The proportion of all U.S. teens who believe the effect of social media has been neutral on people their age is 0.42 with a margin of error of 0.036. Estimate the range of all U.S. teens who believe the effect of social media has been neutral on people their age.

2. In a survey of 175 students at Satellite High School (SHS), 82 reported that they use Facebook.

- a. The estimated proportion of all Satellite High School students who use Facebook is _____ with a margin of error of 0.074.
- b. The estimated proportion of all Satellite High School students who use Facebook is between _____ and _____.
- c. Do you think it's reasonable to conclude that the majority of SHS students use Facebook? Explain your reasoning.
- d. Suppose there are 1,385 students enrolled at Satellite High School. The estimated number of all Satellite High School students who use Facebook is between _____ and _____.



3. A large company tracked the amount spent on each of their Amazon purchases over the past four years. For a sample of 135 purchases, the average amount spent was \$35.23 with a margin of error of $\pm \$2.16$.

If the company predicts they will make 50 purchases on Amazon in the next month, what is the range of the predicted total amount they will spend?



12.6.4 Your Turn!

1. A new drug was given to a random sample of 400 people with acne. The results showed that 210 of the people's acne improved and the results had a margin of error of ± 0.049 .
- a. Based on these results, can it be said that more than half the people with acne will improve if they are given the drug?

- b. If the new medication will be administered to 5,552 people, how many are expected to see improvement?

Between _____ and _____
are predicted to see an improvement in their acne.

2. During harvesting season at Florida Blue Farms, hand pickers collected 200 pounds of blueberries and found a 95% pack-out rate. The margin of error for the sample was $\pm 3\%$. The pack-out rate is the percentage of collected blueberries that can be packaged to be sold, based on Florida Blue Farms' quality standards.
- The estimated percentage of all blueberries that are hand-picked and can be packaged and sold is between _____ and _____.
 - If 2,800 blueberries are hand-picked, what is the most that are expected to be packaged and sold?
3. Publix Supermarkets Inc., commonly known as Publix, has 831 locations in Florida. Publix offers free cookies to children to eat while they're in the store. A survey of 100 store managers found that the average number of cookies given out on a weekday was 15.5, with a margin of error of 2.1.

Complete the statement.

The estimated total number of free cookies given out each weekday for all Florida locations is between _____ and _____.



12.6.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each.



I need help.



I got it.



I could help someone.

Classifying data as categorical or quantitative, and as univariate or bivariate			
Creating and interpreting graphical displays for categorical data			
Creating and interpreting graphical displays for numerical data			
Explaining the difference between correlation and causation			
Using the margin of error to estimate population totals, means, and percentages			

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